

T5.3 Agentic Wrapper Framework

Implementation Summary — M6

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T5.3 is a single Python container that serves all four VIGILANCE GA pilots simultaneously (OTE/TELECOM · Siemens/INDUSTRY_4 · Rotterdam/MARITIME · CaixaBank/FINANCE). It is the operational bridge between raw security events and the cybersecurity tools that respond to them. Pilot detection happens per-event in C1 — no restart or reconfiguration is needed to switch between pilots.

Architecture — 6 Components

C1 — Event Ingestion & Normalization

Converts any raw security event into a structured **CanonicalEvent**. Five parsers tried in priority order:

Parser	Handles	Pilot detection
OTJsonParser	Siemens OPC-UA / Modbus JSON	Hard-wires INDUSTRY_4
CEFParse	ArcSight Common Event Format	From CEF fields
ECSParse	Elastic Common Schema	From agent.type keywords
SyslogParse	RFC 3164 syslog	Emits UNKNOWN
LLMParse	Fallback — any unknown format (mistral:7b)	LLM extraction, validated

The LLM is explicitly instructed never to guess the pilot — returns null for ambiguous events. UNKNOWN pilot falls back to TELECOM profile with a warning log.

CanonicalEvent fields: core (event_id, type, severity, src_ip, target, pilot, timestamp) + pilot-specific extensions: telecom (subscriber_id, imsi, cell_id), OT (plc_id, scada_zone, ot_protocol, ot_safety_flag), maritime (vessel_id, port_zone, ais_mmsi), finance (account_id, transaction_id, fraud_score).

C2 — Agentic Interaction Layer (mistral-nemo)

Multi-turn tool-calling loop (up to 5 turns). For each event: receives CanonicalEvent + sector LLM prompt → may call tools → produces **AgentDecision** (threat type, recommended actions, confidence score).

Tool	Parameters
query_siem_logs	target: str, window_min: int
query_iam_sessions	target: str
query_threat_intel	ioc: str

Pilot	Available actions
TELECOM	block_ip · revoke_session · notify_soc · update_acl

Pilot	Available actions
INDUSTRY_4	isolate_plc · revoke_ot_session · notify_soc · update_zt_policy
MARITIME	block_vessel_access · quarantine_cargo_system · notify_port_authority · notify_soc
FINANCE	freeze_account · block_transaction · notify_fraud_team · escalate_to_compliance · notify_soc

C3 — Action & Policy Execution (mistral-nemo)

- **PolicyTranslator**: translates natural-language policy_update into OPA/Rego rule using mistral-nemo. In INTEGRATED mode, Rego rule published to t53.policy_updates for T5.5.
- **ActionExecutor**: in STANDALONE mode, dispatches each action to the appropriate C4 adapter. Always passes mode=safe-state for isolate_plc actions.

C4 — Tool Adapter Layer (12 plugins, currently stubs)

Pilot	Plugins	Actions
TELECOM (OTE)	SIEMPlugin · IAMPlugin · IDSPlugin	block_ip · revoke_session · notify_soc · update_acl
INDUSTRY_4 (Siemens)	SIEMPlugin · IAMPlugin · SCADAPPlugin	isolate_plc (safe-state enforced) · revoke_ot_session · notify_soc · update_zt_policy
MARITIME (Rotterdam)	PortSIEMPlugin · PortIAMPlugin · PortOpsPlugin	block_vessel_access · quarantine_cargo_system · notify_port_authority · notify_soc
FINANCE (CaixaBank)	BankSIEMPlugin · BankIAMPlugin · FraudEnginePlugin	freeze_account · block_transaction · notify_fraud_team · escalate_to_compliance · notify_soc

SCADAPPlugin enforces safe-state-first: isolate_plc raises ValueError if mode=safe-state is not passed. C3 always passes it automatically. All adapters are currently simulation stubs — real API calls are planned M10–M15.

C5 — Safety, Audit & Simulation

SafetyGate — four rule-based checks before any action executes:

Check	Rule	Fail result
1 — Confidence	agent_confidence ≥ pilot threshold (0.80 / 0.85 FINANCE)	ESCALATE
2 — Protected IP	src_ip not in pilot protected_ranges	ESCALATE
3 — Proportionality	action count ≤ 5	REJECTED
4 — OT scope	policy_update must not target full OT network / all zones	ESCALATE

ESCALATE cases trigger a semantic second-opinion call to mistral:7b → decides APPROVE or REJECT. Final verdicts: APPROVED · REJECTED · ESCALATE.

AuditLog — immutable per-event records with pilot-prefixed IDs (aud-OTE-* · aud-SIE-* · aud-ROT-* · aud-CAI-*). Stores guardrail verdict, all action results, timestamps.

SimulationMode — dry_run (log everything, execute nothing) and digital_twin (also subscribes to dt.events.synthetic for WP3 D-VISOR synthetic events).

C6 — Sector Profile Manager

Loads all four YAML profiles at startup. Each SectorProfile carries: pilot identifier, tool plugin list, policy templates, LLM system prompt (sector-specific, injected into C2), ot_safety_flag, confidence_threshold, and protected_ranges. Per-event routing via _profile_for(pilot) and _adapters_for(sector).

Pipeline Modes

Mode	Env var	Flow
STANDALONE (default)	VIGILANCE_MODE=STANDALONE	Full C1→C2→C5→C3→C4 inside T5.3. C2 uses Mistral Nemo internally. CanonicalEvent also published to t53.canonical_events for observability.
INTEGRATED	VIGILANCE_MODE=INTEGRATED	Two threads, independent pika connections. Thread 1: raw event → C1 → t53.canonical_events → T5.4. Thread 2: t53.action_requests → C5+C3+C4 → t53.policy_updates + t53.actions.dispatch + t53.results.
DIGITAL_TWIN	VIGILANCE_MODE=DIGITAL_TWIN	Same as STANDALONE plus third consumer on dt.events.synthetic for WP3 D-VISOR injection.

Broker Topics (all pre-declared in RabbitMQ)

Topic	Dir	Description
pilot.events.raw	IN	Raw events from pilot environments — entry point
t53.canonical_events	OUT	C1 output — T5.4 input in INTEGRATED mode; observability in STANDALONE
t53.action_requests	IN	T5.4 output — triggers C5+C3+C4 in INTEGRATED mode
t53.policy_updates	OUT	C3 Rego rules → T5.5 ZTA blueprint refinement (fire-and-forget)
t53.actions.dispatch	OUT	C4 action requests → pilot tools (fire-and-forget)
t53.results	OUT	Final ExecutionResult after every event
dt.events.synthetic	IN	WP3 D-VISOR synthetic events for digital twin mode

REST API — T5.6 Integration Point (port 8000)

Method	Endpoint	Description
GET	/api/v1/health	Liveness check — returns loaded pilots and mode
GET	/api/v1/profiles	All four sector profiles with thresholds and plugin lists
POST	/api/v1/events	Submit raw event — returns ExecutionResult (STANDALONE) or 202 (INTEGRATED)
POST	/api/v1/action-requests	Submit ActionRequest for C5+C3+C4 execution
GET	/api/docs	Swagger UI (interactive)
GET	/api/redoc	ReDoc documentation
GET	/api/openapi.json	OpenAPI 3.1 spec

LLM Usage

Model	Component	When	What it does
mistral:7b	C1 LLMParser	Unknown log format	Extracts all CanonicalEvent fields from free-text
mistral:7b	C5 SafetyGate	ESCALATE cases only	Semantic second-opinion: APPROVE or REJECT

Model	Component	When	What it does
mistral-nemo	C2 AgentLoop	Every event	Multi-turn reasoning — tool calls + threat/action decision
mistral-nemo	C3 PolicyTranslator	When policy_update set	NL → OPA/Rego policy rule translation

Both models served by Ollama (local Docker container). When OLLAMA_BASE_URL is unset, StubLLMProvider is used automatically — deterministic responses, all 101 tests pass offline.

Infrastructure

Component	Details
RabbitMQ 3.13	All 7 queues pre-declared from infra/rabbitmq/definitions.json at startup
Ollama	Serves mistral:7b (~4 GB) and mistral-nemo (~7 GB) — downloaded once into persistent Docker volume
Docker Compose	4 services: rabbitmq · ollama · ollama-init (one-shot model downloader) · vigilance-gate
vigilance/main.py	Combined endpoint: REST API (uvicorn, port 8000) in daemon thread + broker consumer(s) in main thread

Developer Tools

Tool	Description
tools/publish_event.sh	Sends any raw event to pilot.events.raw via RabbitMQ Management API. Supports CEF, JSON, syslog for all 4 pilots.
tools/simulate_t54.sh	Simulates T5.4: auto-consumes a CanonicalEvent, derives sector-appropriate actions, publishes ActionRequest to t53.action_requests.

Test Suite — 101 Tests, All Passing (offline)

Suite	Tests
test_c1_ingestion.py	All 5 parsers + normalizer
test_c2_agentic.py	AgentLoop tool calls, final decision, fallback
test_c3_execution.py	ActionExecutor, PolicyTranslator, unknown action
test_c4_adapters.py	All 12 plugins including SCADA safe-state enforcement
test_c5_safety.py	SafetyGate verdicts, AuditLog, SimulationMode
test_c6_profiles.py	All 4 profile loads, caching, validation
test_api.py	All REST endpoints, 4 pilots, Swagger/ReDoc/OpenAPI
tests/scenarios/ (4 files)	Full end-to-end: OTE · Siemens · Rotterdam · CaixaBank

Not Yet Implemented — Planned M7–M18

Item	Current state	Planned
Real C4 adapters (live tool APIs)	Stubs — simulate responses	M10–M15 per pilot

Item	Current state	Planned
T5.1 RAG in C2	Interface not yet wired	M10–M12
T5.2 agent selection	Uses Mistral Nemo directly	M13–M15
Real T5.4 orchestration	Simulated via simulate_t54.sh	M12–M13
API key authentication	Header stub — no enforcement	M7–M9
Audit REST endpoint	Not yet implemented	M7–M9
Adaptive confidence thresholds	Not yet	M16–M18
Long-term memory for C2	Not yet	M16–M18